Supplementary Information

Academic Social Media Migration Through a Psychology of Science Lens

Table of Contents

Table S1 2

Table S2 3

Figure S1 4

Figure S2 5

Supplementary Appendix 1: robustness of results 6

Supplementary Appendix 2: overview of data processing steps 13

Supplementary Appendix 3: preregistration deviations 17

Supplementary Appendix 4: number of sources smooth term 24

Supplementary Appendix 5: could widely followed individuals drive migration? 25

Supplementary Appendix 6: regional differences in migration rates 28

Supplementary Appendix 7: context of the open science terms in the profile 32

Supplementary Appendix 8: Return to Twitter after the migration period 33

References 35

Family	Link Function	Formula	Adjusted R2	Deviance Explained
binomial	logit	Migration ~ s(Social Impact) + Open Science Endorsement + s(Twitter Account Age) + s(Twitter Tweet Count)	0.247	19.80%
Parametric Coefficients				
	Estimate	**Std. Error**	**z Value**	**Pr(>|z|)**
(Intercept)	-0.02805	0.03332	-0.842	0.4
Open Science Endorsement	2.30044	0.29071	7.913	2.51e-15 ***
Approximate significance of smooth terms:				
	edf	**Ref. df**	**Chi.sq**	**p-value**
s(Social Impact)	8.573	8.895	599.34	<2e-16 ***
s(Twitter Account Age)	8.369	8.89	56.08	<2e-16 ***
s(Twitter Tweet Count)	7.205	7.771	168.88	<2e-16 ***

Table S1. GAM results for social impact model (training set). This table presents the summary of the Generalized Additive Model (GAM) for the social impact model used in the study. Although our primary focus is on the machine learning (predictive) performance of the model, we include this GAM output for completeness. The “Parametric coefficients” section shows the estimated intercept and the coefficient for open science endorsement, while the “Smooth terms” section lists the effective degrees of freedom (EDF) and associated statistics for the smooth functions of the standardized predictors (social impact, twitter account age, and twitter tweet count). Larger EDF values indicate more flexible smooths. R-sq.(adj), deviance explained, and UBRE (0.12) quantify overall model fit, and n reports the sample size used (6384). Statistical significance indicates evidence that the corresponding term contributes to explaining the outcome. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Family	Link Function	Formula	Adjusted R2	Deviance Explained
binomial	logit	Migration ~ s(Social Impact) + Open Science Endorsement + s(Twitter Account Age) + s(Twitter Tweet Count)	0.249	20%
Parametric Coefficients				
	Estimate	**Std. Error**	**z Value**	**Pr(>|z|)**
(Intercept)	-0.02970	0.02841	-1.046	0.296
Open Science Endorsement	2.41767	0.26448	9.141	<2e-16 ***
Approximate significance of smooth terms:				
	edf	**Ref. df**	**Chi.sq**	**p-value**
s(Social Impact)	8.823	8.976	765.81	<2e-16 ***
s(Twitter Account Age)	8.423	8.907	71.58	<2e-16 ***
s(Twitter Tweet Count)	7.340	7.936	219.18	<2e-16 ***

Table S2. GAM results for social impact model (complete dataset). This table presents the summary of the Generalized Additive Model (GAM) trained on the complete downsampled dataset (i.e., both the training and test sets pooled). While our primary analyses focus on predictive performance based on the original train–test split, we fit this "full-data" model to obtain a more stable estimate of effect size for Open Science Endorsement. Accordingly, the odds ratio reported in the main paper is derived from this model's parametric coefficient. The "Parametric coefficients" section shows the estimated intercept and the coefficient for open science endorsement, while the "Smooth terms" section lists the effective degrees of freedom (EDF) and associated statistics for the smooth functions of the standardized predictors (social impact, twitter account age, and twitter tweet count). Larger EDF values indicate more flexible smooths. R-sq.(adj), deviance explained, and UBRE (0.116) provide measures of overall model fit, and n reports the sample size used (7980). Statistical significance indicates evidence that the corresponding term contributes to explaining the outcome. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

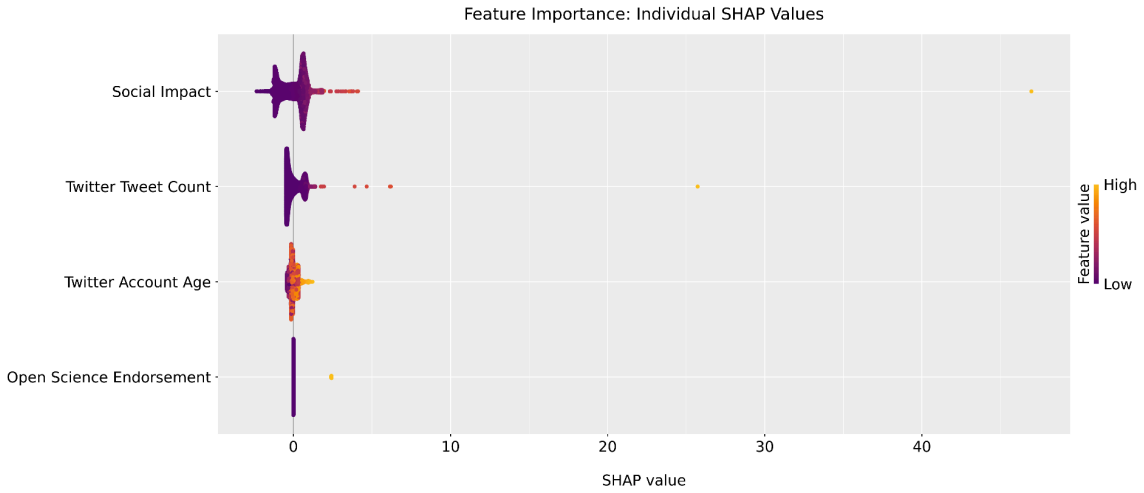


Figure S1. Feature importance for social impact model (beeswarm plot; complete dataset). This figure displays per-instance SHAP values (on the x-axis) for each feature in the social impact model, illustrating how each feature influences individual predictions. Each dot represents a single observation, and the color corresponds to the relative value of that feature for the observation. Features with wider spreads on the x-axis have a greater overall impact on the model's output. We use the complete downsampled dataset (i.e., both the training and test sets pooled) here for more stable estimates of feature importance.

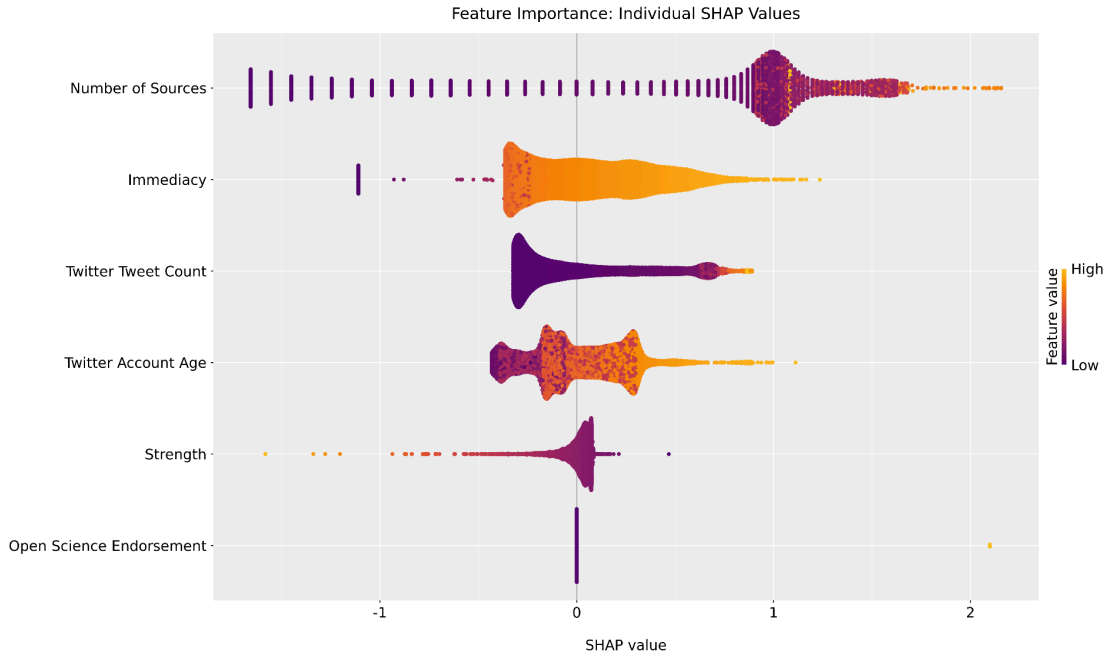


Figure S2. Feature importance for social impact components model (beeswarm plot; complete dataset). This figure displays per-instance SHAP values (on the x-axis) for each feature in the social impact components model, illustrating how each feature influences individual predictions. Each dot represents a single observation, and the color corresponds to the relative value of that feature for the observation. Features with wider spreads on the x-axis have a greater overall impact on the model's output. We use the complete downsampled dataset (i.e., both the training and test sets pooled) here for more stable estimates of feature importance.

Supplementary Appendix 1: robustness of results

The analysis reported here was performed to ensure that the modeling results are robust for different subsets of the data and train-test splits. For completeness, we also include the logistic regression analysis that was originally planned, but not used as the necessary assumptions were violated.

Method

As stated in the manuscript, we addressed severe class imbalance in the outcome variable by applying random undersampling to the majority class ("not migrated"), reducing its size to match that of the minority class ("migrated") (Japkowicz, 2000). Subsequently, 20% of the data was reserved for the test set. To examine whether the results are robust, we repeated the undersampling process for 50 different iterations, creating distinct majority-class downsamples and train-test splits.

To assess the stability of feature importance using SHAP, we pooled the entire downsampled dataset (both training and test set) on each iteration to obtain more robust estimates of feature importance. Similarly, we used the entire downsampled dataset to assess the stability of the odds ratio for the Open Science Endorsement predictor.

Results

As can be seen in Figure SA1.1 and Figure SA1.2, the predictive performance of the social impact model and the social impact components model shows low variability across different random seeds with different downsamples and train–test splits. It can thus be considered stable.

The feature importance, as assessed using SHAP, also shows low variability for both models (see Figure SA1.3 and Figure SA1.4), suggesting that it is robust. That is, the interpretation of what features are most important is not affected by the seed. Furthermore, the odds ratio for the Open Science predictor consistently exceeded one, indicating higher

odds of migrating than not migrating. Notably, the confidence intervals never included one (see Figure SA1.5).

Finally, the logistic regression analysis revealed statistically significant effects for all predictors and covariates (see Table SA1.1).

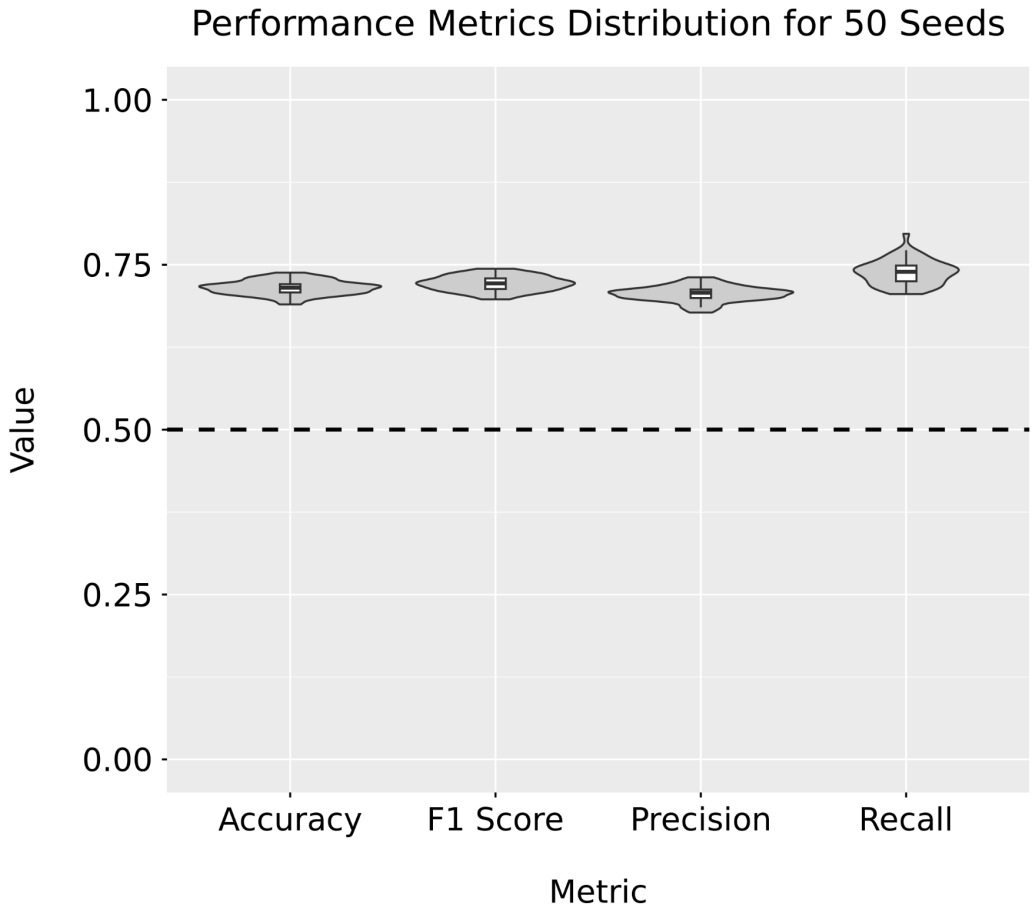


Figure SA1.1. Distribution of predictive performance for the social impact model. These violin plots, overlaid with boxplots, illustrate how the model's test-set predictive performance varies across 50 different random seeds (different downsamples and train–test splits). The width of each violin reflects the density of performance values, while the boxplot inside each violin highlights the median and interquartile range.

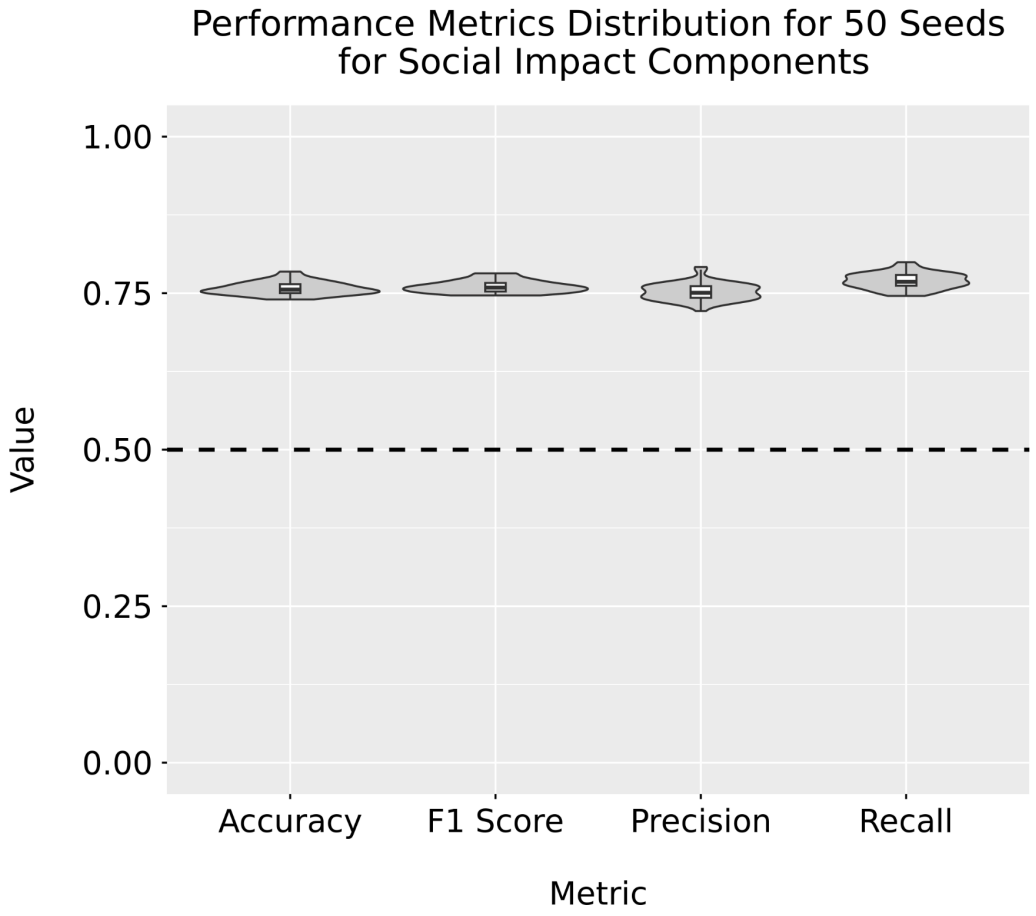


Figure SA1.2. Distribution of predictive performance for the social impact components model. These violin plots, overlaid with boxplots, illustrate how the model's test-set predictive performance varies across 50 different random seeds (different downsamples and train–test splits). The width of each violin reflects the density of performance values, while the boxplot inside each violin highlights the median and interquartile range.

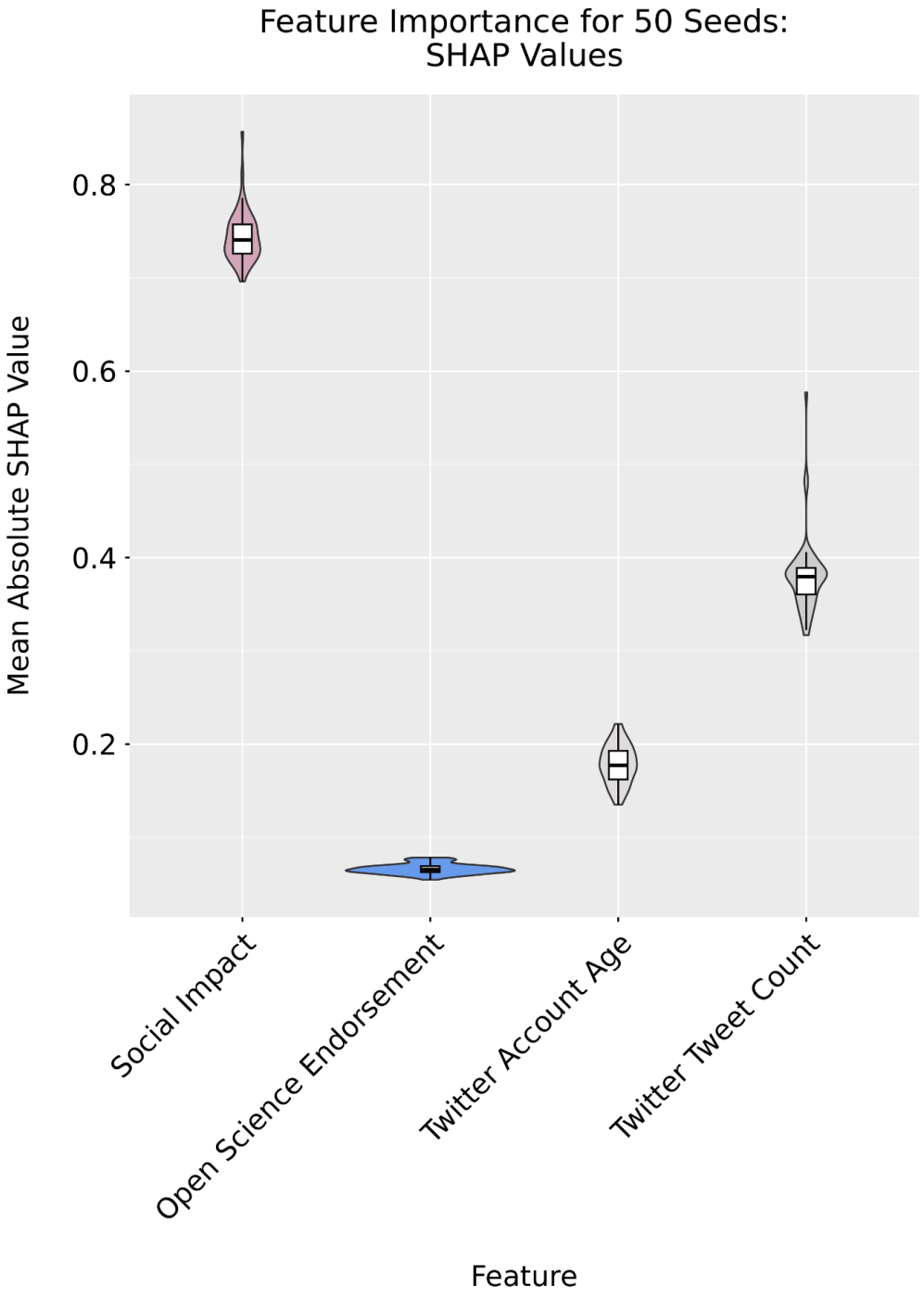


Figure SA1.3. Distribution of SHAP values for feature importance for the social impact model (complete dataset). These violin plots, overlaid with boxplots, depict how each feature's SHAP values vary across 50 different random seeds (different downsamples). Wider distributions within the violins indicate greater variability in a feature's contribution to the model's predictions, while the boxplots highlight the median and interquartile range. We use the complete downsampled dataset (i.e., both the training and test sets pooled) here for more stable estimates of feature importance.

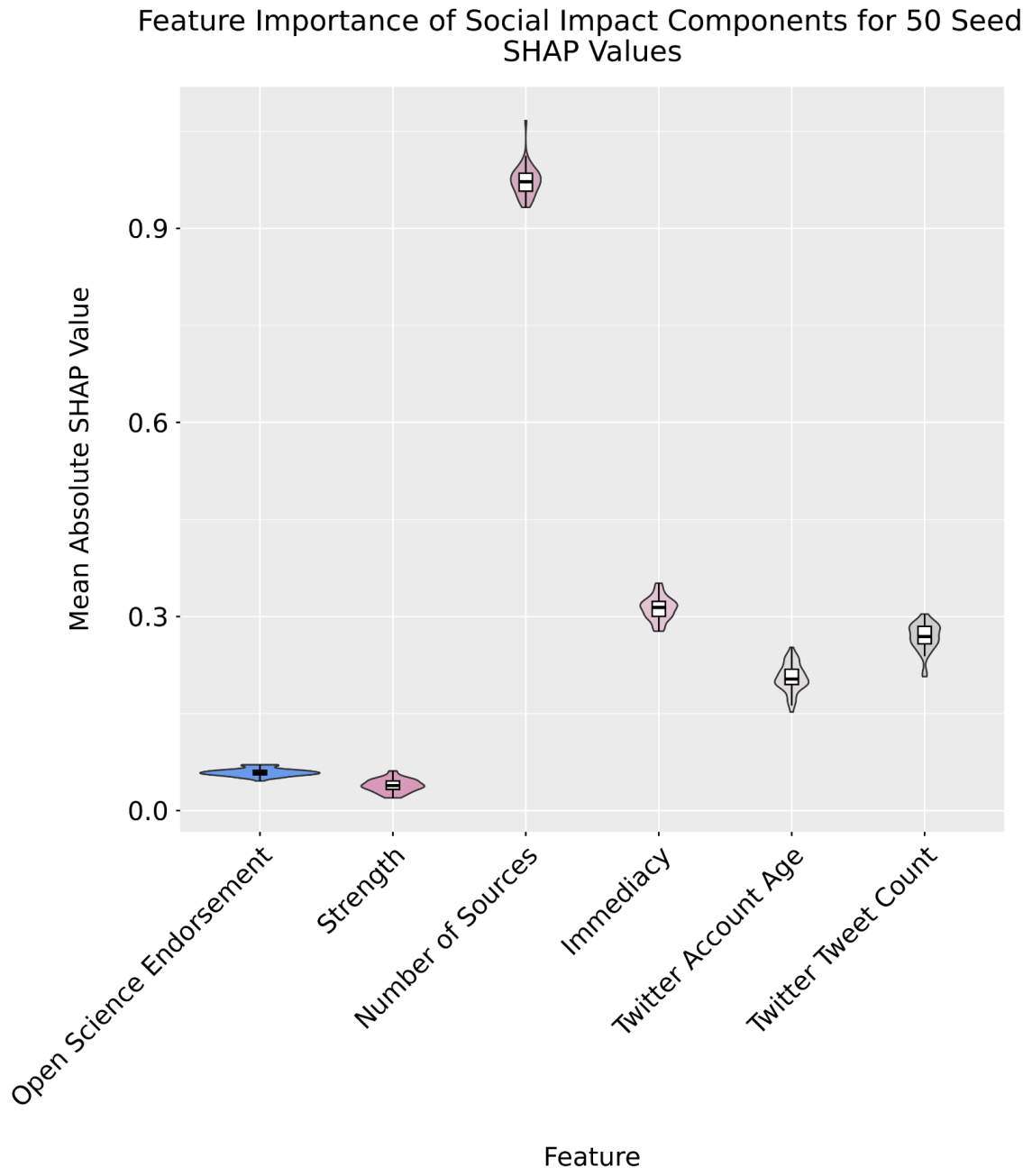


Figure SA1.4. Distribution of SHAP values for feature importance for the social impact components model (complete dataset). These violin plots, overlaid with boxplots, depict how each feature's SHAP values vary across 50 different random seeds (different downsamples). Wider distributions within the violins indicate greater variability in a feature's contribution to the model's predictions, while the boxplots highlight the median and interquartile range. We use the complete downsampled dataset (i.e., both the training and test sets pooled) here for more stable estimates of feature importance.

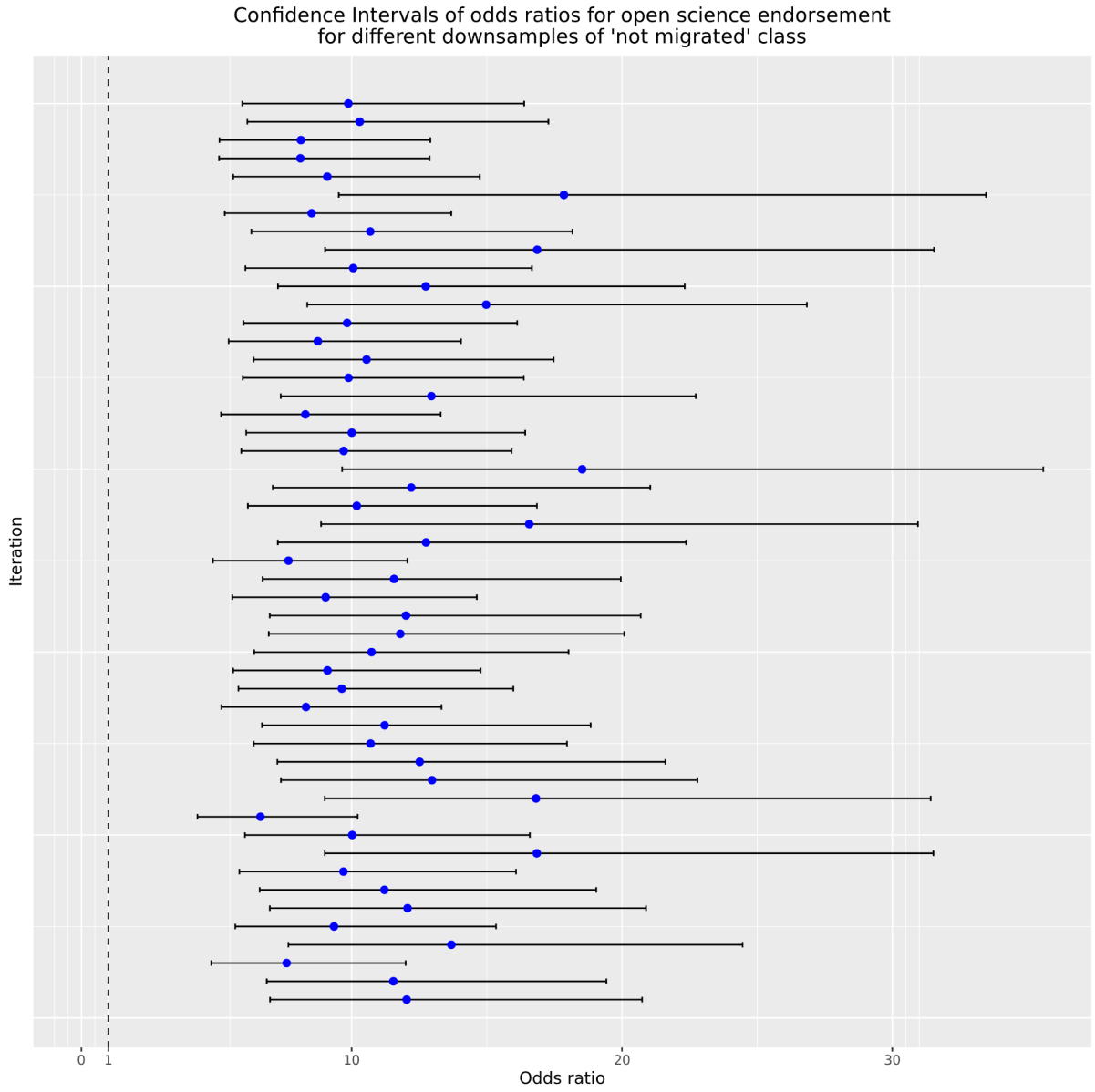


Figure SA1.5. Distribution of odds ratios for the social impact model (complete dataset). Blue dots and error bars show estimated odds ratios and their 95% confidence intervals across 50 different random seeds (different downsamples). The dashed line at an odds ratio of 1 represents the point where the odds of migrating equals the odds of not migrating. Notably, all 50 estimates lie above this line, indicating consistently higher odds of migrating.

Formula	Family	Null Deviance (df)	Residual Deviance (df)	AIC
Migrated ~ Social Impact + Open Science Endorsement + Twitter Account Age + Twitter Tweet Count	binomial	41201 (258408)	38089 (258404)	38099

Coefficients

	Estimate	Std. Error	z Value	Pr(>|z|)
(Intercept)	-4.410044	0.018849	-233.968	<2e-16 ***
Social Impact	0.363945	0.008776	41.468	<2e-16 ***
Open Science Endorsement	2.225362	0.081422	27.331	<2e-16 ***
Twitter Account Age	0.348953	0.017863	19.535	<2e-16 ***
Twitter Tweet Count	0.066520	0.008065	8.248	<2e-16 ***

Table SA1.1. Logistic regression results. For completeness, this table shows the results of logistic regression that was originally planned, but ultimately not used because the necessary assumptions were violated. Using the entire dataset (N = 258,409), the regression models whether a user migrated using the predictors open science endorsement, and standardized social impact, and the standardized covariates Twitter account age and Twitter tweet count. Reported are the coefficient estimates (Estimate), standard errors (Std. Error), test statistics (z value), and p-values. The model deviance and AIC values indicate the overall fit of the logistic regression model.
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Supplementary Appendix 2: overview of data processing steps

This appendix provides an overview of the data processing steps of the study. At the core, following data collection, the Twitter researcher profiles are used as a foundation that is enriched with various additional information (e.g. on Twitter and Mastodon post activity in the time period of interest, social impact, and open science endorsement). Ultimately, this dataframe is subjected to the analysis, where each row is a person (researcher) who either migrated or not (outcome variable; column), and the columns hold the features. For further details and exact implementation, please refer to the code and preprocessed data.

1. Initial data collection
    - Web Scrape OpenCheck list
    - Twitter data collection
        - Twitter profiles of researchers
        - Tweets from researchers during Nov-Dec 2022 (initial migration period)
        - Tweets from researchers during Apr 2023 (return period)
        - Migration-related tweets for influencer identification
2. Initial processing (before Mastodon data collection)
    - Identify TwitterMigration influencers
        - Subset of original (not reposts) migration-related tweets by those researchers found in Twitter profiles
        - Influencers are researchers with at least one original migration-related tweet
        - Calculate influence score for each influencer: weighted sum of global (20%) and local (80%) metrics
    - Manually review top 100 influencers
        - Remove those who do not qualify as researchers or relevant influencers based on their profile description, their top ten most influential tweets, and/or web search results.
        - Account for anti-migration influencers: Weigh influence score (-1/1*X) using top 10 most influential tweets (person against migration if >= 1 post against migration)
    - Collect influencer followers on Twitter
    - Extract Mastodon handles from researcher Twitter profiles
        - Extract valid handles from known Mastodon instances

        - Manually check instance rules of the 125 most common instances (representing 95% of the users found in these handles), ensuring that data collection is not prohibited
    - Mastodon data collection
        - Mastodon profiles created or reactivated in initial migration period
        - Mastodon posts created in initial migration period
3. Main processing (data collection completed)
    - Remove some researchers from Twitter profiles
        - Remove Twitter profiles with missing Twitter timelines as it is not possible to determine activity and migration status (Note: influencers without timelines are kept until social impact calculation completed)
        - Remove Twitter profiles with missing Mastodon timelines (for people with Mastodon account). (Note: influencers without timelines are kept until social impact calculation completed)
        - Remove Twitter profiles with Mastodon bot accounts
        - Remove Twitter profiles created after the start date (Nov 1st 2022).
    - Determine Mastodon activity
        - Sum of toots, reposts etc. for each person with an account created or revitalized in Nov-Dec for Nov-Dec 2022
    - Determine Twitter activity
        - Sum of tweets, retweets etc. from Mastodon account creation (or revitalization) date to end of Dec 2022
    - Determine migration status: compare Twitter and Mastodon activity
    - Construct and preprocess Mastodon timeline dataframe for immediacy calculation
    - Get followed influencers for each researcher
    - Calculate Immediacy of researchers with their followed influencers
        - Goal: For each researcher, get relevant profile description and posts in order to determine similarity of research interests with followed influencers based on text
        - Pre-post split
            - For each researcher, use Twitter posts from Nov 1st 2022 until Mastodon account creation date and then either Twitter or Mastodon posts based on which platform the user was more active on. Use Twitter or Mastodon profile description (based on migration) as the first post.
            - Remove people from Twitter profiles and pre-post split who have no valid profile description and no posts
            - Remove posts without text, reformat
        - Precompute embeddings for posts
        - Identify research-related posts
            - Cosine similarity of 0.5 with at least one research topic from Open Alex list of research topics

                - Map each post to only those Open Alex research disciplines labels for which cosine similarity threshold is exceeded
                - Remove posts that don't exceed the threshold
            - Run zero-shot BERTopic on the documents
            - Get document-topic probabilities per person per post, and average them to obtain a single document-topic probability vector per person
            - Remove people from profiles who don't have any research-related posts or profiles description
            - Remove followed influencers IDs of influencers that are not contained in the profiles anymore.
            - Calculate immediacy (i.e., similarity of interests) of each user and the followed influencers
                - Mean correlation of mean document-topic probabilities (i.e., across all posts) of the user with probabilities of followed influencers.
    - Calculate strength and number of sources
        - Strength is average influence score of the followed influencers
    - Final social impact score
        - Min-max scale strength, number of sources, and immediacy
            - For strength, we take the absolute values and consider the sign (anti-migration influencers) only after min-max scaling, so that 0 strength stays 0.
            - For immediacy, we do not consider the sign as conceptually it does not make sense for interest in different topics to result in a final negative social impact score when calculating the product of the components.
            - People who don't have any followed influencers have zero strength, immediacy and number of sources
            - Calculate final social impact score
            - Remove influencers without Twitter timelines from profiles
            - Remove influencers without Mastodon timelines from profiles
    - Flag open science endorsement
        - Look for open science patterns in the profile description (either Twitter or Mastodon profile, depending on migration status)
4. Analysis
    - Z-standardize predictors: the social impact score, Twitter account age, and Twitter tweet count; for the social impact components model, the min-max scaled strength, immediacy, and number of sources. Influence scores of individual influencers (Table 1) are not standardized.
    - Predictive performance on test set for social impact and social impact components model
        - Downsample the majority class to the size of the minority class
        - Train GAM on 80% of the data, evaluating performance on the remaining 20% of the data (test set).

- SHAP values for feature importance for social impact and social impact components model
    - Calculate SHAP on complete downsampled dataset (i.e., both the training and test sets pooled).
    - For calculating SHAP values for the social impact components model, extreme outliers (above or below 10) in the predictors are clipped.
- Odds ratio for Open Science Endorsement for social impact and social impact components model
    - Determine odds ratio on complete downsampled dataset (i.e., both the training and test sets pooled)
    - Residual diagnostics are performed to check if the model is fitting well.
- Check robustness of predictive performance for social impact and social impact components model
    - Calculate performance on test set for 50 different seeds
    - Rationale: ensure that the predictive performance is robust for different subsets of the data (downsamples) and distinct train-test splits
- Check robustness of SHAP values for feature importance for social impact and social impact components model
    - Calculate SHAP values for 50 different seeds and distinct train-test splits
- Number of sources: predicted probabilities (Figure 3) for social impact components model
    - The social impact components model is refit with the raw (non-standardized) and easily interpretable number of sources predictor along with the other standardized predictors. Predicted migration probabilities are computed with the other predictors at their full-sample means and open science endorsement set to "no", corrected for the class-balanced sample (King & Zeng, 2001). The smooth term itself is shown in Supplementary Appendix 4.
    - Extreme outliers in the raw number of sources predictor are clipped similar to outliers in the other predictors.
- Logistic regression
    - For completeness, calculate logistic regression analysis which was initially planned, but ultimately not used because the necessary assumptions were violated.

Supplementary Appendix 3: preregistration deviations

Table SA3.1 shows the preregistration deviations. For each deviation, the original wording, a deviation description as well as reader impact is provided.

Table S1

Preregistration Deviations Table (adopted from Willroth & Atherton, 2024)

Deviations

#	Details		Original Wording	Deviation Description	Reader Impact
1	Type	Sample	"First, we will make use of open lists of Mastodon accounts (that are already matched with Twitter accounts; see M2). Second, we will check for Mastodon account handles in Twitter profiles and screen names (La Cava, 2023). The sum of unique Mastodon accounts will be compared to the above-mentioned estimation (i.e., 16,600 accounts) in order to decide whether a third source for finding Mastodon account names might be necessary. In this case, we will use the following approach to identify additional researchers on Mastodon: Researchers on Mastodon will be identified using ground truth data sources (see M2) and network crawling	Automatic crawling proved to be legally problematic due to the different terms of service of Mastodon instances. Hence, we used only open lists and account handles.	The number of Mastodon accounts as reported in our study may not reflect the true number of researchers with accounts on both platforms. However, this does not affect our hypothesis testing.
	Reason	Plan not p…			
	Timing	Before dat…			

			using a chain-referral sampling algorithm (as used in Müller et al., in press)" (AP3)*		
2	Type	Analysis	"We will use a logistic regression model for both hypotheses" (AP6)	Our data yielded violations of model assumptions. Specifically, the assumption of linearity of the independent variables with the log-odds was violated. The model did not fit well (as indicated by the residuals). Hence, we opted for generalized additive models (GAMs), which can handle this type of data.	We report machine learning based model performance and values (SHAP) instead of inference statistics. For the sake of completeness, results of logistic regression are provided in the supplements.
	Reason	Plan not p…			
3	Type	Hypotheses	"H1: Researchers who are under higher social influence from #TwitterMigration influencers are more likely to migrate to Mastodon than researchers who are under lower social influence." "H2: Researchers who endorse the open science movement are more likely to migrate to Mastodon than researchers who do not."	We noticed that the wording of H1 may be misleading by implying a group comparison. In addition, we Erroneously used the term "social influence", although the hypothesis is derived from social impact theory. Also, social impact is the main outcome measure. In addition, we dismissed the use of a hashtag for better readability. Upon reviewer feedback highlighting the inherent	The updated wording of H1 should avoid a misleading implication of group comparison and causal effects, which are not reflected in our observational design and analytical procedure using generalized additive models. The updated wording of H2 should make it clear that it is about open science itself, not the movement.
	Reason	Miscomm…			
	Timing	After data …			

				confound between social influence and homophily in observational network data, we revised H1's wording to avoid implying a causal pathway from specific influencers that our design cannot establish. Accordingly, we rephrased as follows: "H1: Researchers with higher social impact scores are more likely to migrate to Mastodon." Regarding H2, we dismissed the term “movement” to stress that we focus on the principles of open science, not the movement: “H2: Researchers who endorse open science are more likely to migrate to Mastodon than researchers who do not.”	
4	Type	Variables	“For Immediacy (proximity of influencing source), we focus on social proximity in terms of similar interests.	We encountered a low usage of hashtags, and the hashtags contained little information	The immediacy in terms of similar academic interests is more robust using science topics instead of all
	Reason	Plan not p…			

	Timing	After data …	Specifically, for each user, we will: 1. Extract hashtags*: Extract the max. top 100 most frequently used hashtags for each user (excluding hashtags related to #TwitterMigration, see M1). 2. Calculate embeddings: Use a transformer model (i.e., RoBERTa) to calculate embeddings for each of these hashtags for each user. 3. Weight the embeddings: Multiply the embeddings by their corresponding hashtag frequency to give more weight to the hashtags that a user uses more often. 4. Aggregate embeddings: For each user, average the weighted embeddings of their top 100 hashtags to get a single vector that represents the user's interests. 5. Calculate similarity values: Using these aggregated vectors, calculate cosine similarity (range 0-1) between each pair of users to get their social immediacy scores. 6. Calculate mean similarity score.	regarding academic interests. We opted for identifying academic interests with zero-shot topic modeling based on OpenAlex topics (van Eck, 2024).	hashtags.

			* To obtain hashtags of the users, we will use the tweets or Mastodon posts collected for November/December (2022) (see M9), depending on the person's main social media outlet." (M12)		
5	Type	Variables	"Users are regarded as endorsing open science if they state "Open Science" or grammatical variants in profile descriptions (without negation), or post on open science with a positive sentiment (in average across their max. top 100 posts)" (M12)	A manual inspection of sentiment analysis results revealed that posts on open science with a positive sentiment did not clearly indicate open science endorsement. As an alternative approach, we tried stance detection, which also yielded low classification performance. Hence, we focused on profile descriptions only.	The focus on researchers' explicit mentioning "open science" in their profile descriptions may not reflect the true number of open science endorsing scientists. However, this does not affect our hypothesis testing.
	Reason	Plan not p…			
	Timing	After data …			
6	Type	Variables	"In order to control for potential confounding effects, we will include several covariates (see below). The covariates will be z-standardized. ● Followers Count ● Following Count ● Account Age ● Tweet Count" (M12)	We observed high correlations for both "followers count" and "following count" with our social impact predictor (especially with the "number of sources" component) and the "tweet count" covariate. While most of the correlations did not exceed our predefined threshold of 0.7 set for the logistic regression, we decided to exclude the covariates followers count and	Only account age and tweet count were included as covariates.
	Reason	Plan not p…			
	Timing	After data …			

				following count from the additive GAM. This was done to ensure the interpretability of SHAP values. In additive models, SHAP values assume feature independence, and including correlated predictors can lead to misattribution of feature importance, thereby distorting the interpretability of the model.	

Unregistered Steps

#	Details		Original Wording	Unregistered Step Description	Reader Impact
1	Type	Analysis	"In addition, we will conduct an exploratory analysis, assessing the unique contributions of social impact factors by including them as separate predictors. We may then also consider exploring interactions between different social impact factors." (AP6)	The number of sources (one of the three factors of social impact) turned out to be the most important feature in predicting migration. Thus, it seemed reasonable to look more closely at the relationship between the number of peers talking about migration and the likelihood of migration.	We provide further insight into the critical mass of peers needed to drive behavior change among researchers.
	Timing	After resul…			
2	Type	Select One			
	Timing	Select One			

3	Type	Select One			
	Timing	Select One			

Note. *The abbreviations in parentheses relate to the items of the PRP-QUANT preregistration template (Bosnjak et al., 2022). Our original preregistration can be found on PsychArchives (https://doi.org/10.23668/psycharchives.13062).

Supplementary Appendix 4: number of sources smooth term

Figure 3 in the main text reports predicted probabilities of migration. This appendix shows the underlying quantity estimated by the model: the smooth term for the number of followed influencers, on the log-odds scale (Figure SA4.1), as returned by mgcv::plot.gam().

Generalized additive models impose an identifiability constraint on each smooth term so that it sums to zero across the observed data. The curve therefore shows how much this predictor contributes relative to its own average, not the absolute log odds of migration; the model intercept and the contributions of all other predictors are excluded. A value of zero indicates that a researcher's number of followed influencers contributes exactly the sample average of this term. This occurs at 18 followed influencers and does not correspond to a 50% probability of migrating. For reference, the median is 8 in the full sample and 20 in the class-balanced sample used to fit the model.

Figure 3 is obtained from this curve by adding the model intercept and the contributions of the remaining predictors (held at their full-sample means, with open science endorsement set to 'no'), applying the inverse logit, and correcting the intercept for the class-balanced downsampling (King & Zeng, 2001). The two figures are therefore monotone transformations of one another: wherever this curve rises, predicted probability rises.

The initial rise is steepest up to approximately 44 followed influencers, where the curve reaches a first local maximum and flattens. The exact location of this maximum depends on the range searched (45 within 0–50, 96 within 0–100), and beyond roughly 100 followed influencers only 7.8% of the class-balanced sample (2.4% of all researchers) is observed, so fluctuations in that region should not be interpreted substantively.

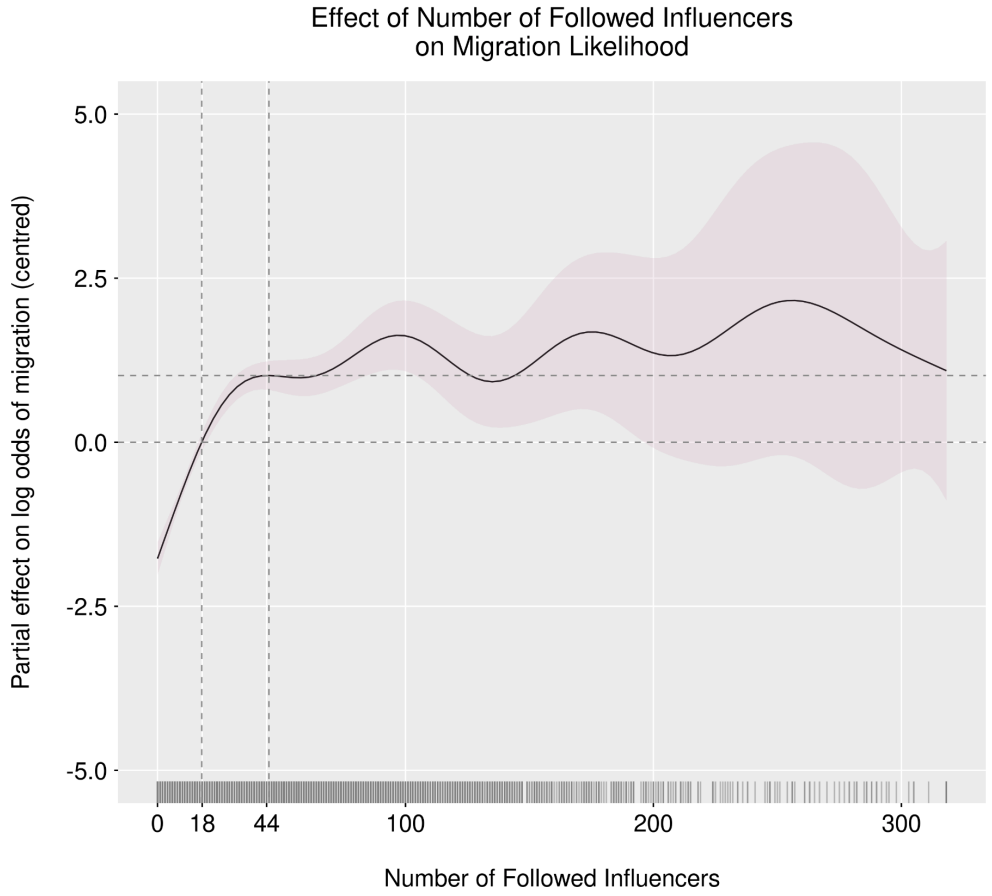


Figure SA4.1. Estimated smooth term from the generalized additive model with the social impact components as separate predictors, on the log-odds scale, plotted over the full observed range. The term is centered so that it sums to zero across the sample; the model intercept and the remaining predictors are not included. The shaded area shows the 95% confidence interval (± 2 SE). Dashed lines mark 18 and 44 followed influencers. The rug marks observed values.

Supplementary Appendix 5: could a few widely followed individuals be driving the migration?

The low contribution of the strength component (Figure 2d) shows that the average influence score of the peers a researcher follows added little to the prediction of migration. It does not by itself rule out a different scenario: that a small number of individuals appear in the networks of most researchers and drive the migration on their own. This appendix

examines that scenario directly using the follow ties between the 258,409 researchers in the analysis sample and the 10,742 potential influencers (4,794,440 ties in total).

Concentration of the follow network

The follow network is diffuse rather than dominated by a few individuals. The median potential influencer is followed by 238 researchers in our sample, the 90th percentile by 976, and the most widely followed individual by 28,380, i.e., 11.0% of all researchers. The ten most-followed influencers together account for only 3.3% of all follow ties, the 100 most-followed for 13.2%, and the 500 most-followed for 31.1% (Table SA5.1). To reach half of the researchers in the sample through at least one followed influencer, the 50 most-followed individuals are required. Migrated researchers are more likely to follow the most widely followed individuals, as expected given that they follow more influencers overall, but even the 20 most-followed individuals are absent from the networks of about a third of migrated researchers. The influence score and the number of followers in our sample are moderately related (Spearman's rho = .60), so the strength component already reflects, in part, whether a researcher follows the most widely followed individuals.

Table SA5.1

Most-followed influencers	Share of all follow ties	Researchers following at least one	Migrated researchers following at least one
Top 1	0.6%	11%	14.9%
Top 5	-	29.7%	45.1%
Top 10	3.3%	36.9%	57.3%
Top 20	5.0%	43.7%	68.2%
Top 50	8.8%	54.6%	82.9%
Top 100	13.2%	-	-

Top 500	31.1%	-	-

Note. N = 258,409 researchers, 10,742 potential influencers, 4,794,440 follow ties. The share of follow ties for the single most-followed influencer is 28,380 / 4,794,440.

Do specific individuals predict migration beyond the social impact components?

Among the followers of each of the 20 most-followed influencers, the migration rate ranged from 2.1% to 6.0%, compared with 1.54% overall. These rates are broadly comparable to what the social impact components model predicts for these same researchers from their own predictor values (2.0% to 5.3%), with deviations in both directions (observed to predicted ratio 0.57 to 1.50, median 1.03), rather than indicating a consistently exceptional effect of particular individuals. To test this formally, we added to the social impact components model (Section 2.4), fitted to the same class-balanced sample as in the main analysis, a binary indicator of whether a researcher follows a given influencer, separately for each of the 20 most-followed influencers. The adjusted odds ratios ranged from 0.43 to 2.25: seven were significantly above one, six significantly below one, and seven did not differ from one ($\alpha = .05$). Following a specific widely followed individual was thus associated with a higher probability of migrating in some cases and a lower probability in others, with no consistent direction. Adding all 20 indicators jointly increased the deviance explained from 28.6% to 30.4% and the out-of-sample AUC on a held-out 20% test set from 0.828 to 0.838.

Summary

The follow network is not concentrated on a few individuals, no single influencer reaches more than 11% of researchers, the migration rate among followers of the most widely followed individuals is consistent with the number of peers they follow, and identifying the most-followed individuals adds only marginally to the predictive performance of the

components model. These results support the interpretation in the main text that migration was related to the number of peers discussing the move rather than to a small set of individual influencers. They do not exclude that specific individuals mattered for specific researchers, which cannot be assessed without complete follower networks (see Limitations).

Supplementary Appendix 6: Regional differences in migration rates

Following reviewer suggestion, this appendix explores whether migration to Mastodon differed by the country and region of researchers' affiliation.

Method

Twitter accounts were linked to OpenAlex author records using the open dataset of paired OpenAlex author IDs and Twitter user IDs by Mongeon et al. (2023; version 2, Mongeon et al., 2024). Of the 258,409 researchers in the final dataset, 236,322 (91.5%) could be matched to an OpenAlex author ID; where a Twitter account was linked to several author IDs, the match to the primary author name was retained. Author records were retrieved from the OpenAlex API for 216,121 of the 234,244 unique author IDs (92.3%); the remaining 7.7% had been retired or merged by OpenAlex since the crosswalk was compiled in February 2024.

For each author, we used the country of the institution listed in their affiliation history for 2022, the year of the migration. Where no 2022 affiliation was on record (21% of authors), the affiliation closest in time was used. In total, a country of affiliation could be assigned to 217,073 researchers, i.e., 84.0% of the final dataset, including 3,402 of the 3,990 migrated researchers (85.3%). Countries were grouped into regions as listed in Table SA6.1; Mexico is grouped with North America, Russia with Europe, and Turkey and Israel with Asia.

Note that this procedure identifies the country of a researcher's institutional affiliation

in 2022 as recorded on their publications, not their nationality. About a quarter of authors with a 2022 affiliation had affiliations in more than one country that year; in these cases, the affiliation whose record extends furthest forward in time was used. A sensitivity analysis using two alternative assignment rules (preferring the affiliation that also covers 2021 and 2023, and excluding authors with affiliations in several countries) changed country-level migration rates by less than one percentage point for all countries with at least 100 researchers, except four countries with fewer than 200 researchers (Iceland, Estonia, Taiwan, Qatar), and left the ranking of the highest-rate countries unchanged.

Results

Migration rates were highest in Europe and lowest in Asia and Latin America (Table SA6.1). Europe accounted for 44% of researchers with an identified country but 54% of migrants; conversely, Asia accounted for 8.6% of researchers but 2.9% of migrants.

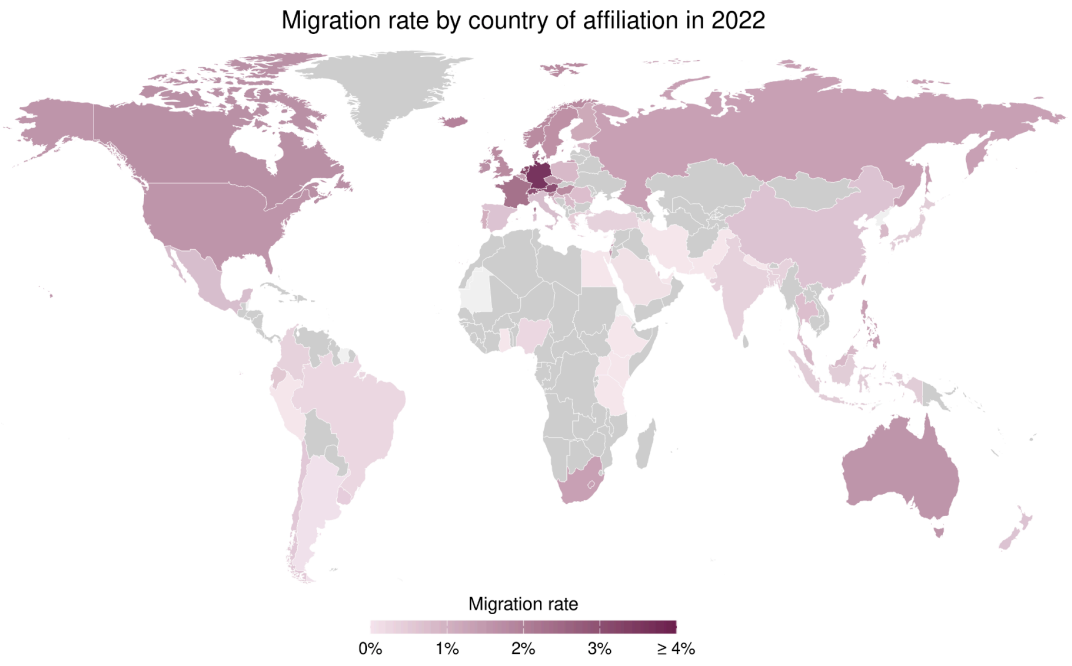


Figure SA6.1. Migration rate by country of affiliation in 2022. Countries with at least 100 identified researchers are coloured by migration rate, with the colour scale capped at 4%; countries with fewer identified researchers are shown in mid grey and countries without affiliation data in light grey.

Table SA6.1. Migration by region of affiliation in 2022

Region	Researchers	Migrated	Migration rate	Share of all migrants	Share of all researchers
Europe	95,952	1,842	1.92%	54.1%	44.2%
North America	78,206	1,233	1.58%	36.2%	36.0%
Oceania	12,441	181	1.45%	5.3%	5.7%
Africa	2,923	17	0.58%	0.5%	1.4%
Asia	18,603	100	0.54%	2.9%	8.6%
Latin America	7,801	26	0.33%	0.8%	3.6%
Other	1,147	3	0.26%	0.1%	0.5%
Total	217,073	3,402	1.57%	100.0%	100.0%

Note. The overall migration rate in the final dataset is 1.54% (3,990 of 258,409); the total row refers to the 217,073 researchers with an identified country, among whom migrants are slightly over-represented (85.3% of migrants vs. 84.0% of all researchers could be assigned a country).

At the country level, the highest migration rates among the 69 countries with at least 100 identified researchers were observed in Germany (3.6%), the Netherlands (3.2%), Austria (3.0%), and Switzerland (3.0%), followed by France and Denmark (Table SA6.2). The United States and the United Kingdom, which together account for almost half of all migrants, had rates close to the overall average (1.6% and 1.8%). Figure SA6.1 shows the migration rate for all countries with at least 100 identified researchers.

Table SA6.2. Countries with the 20 highest migration rates (at least 100 identified researchers)

Country	Researchers	Migrated	Migration rate	Share of all migrants

Germany	9,572	343	3.58%	10.1%
Netherlands	5,324	171	3.21%	5.0%
Austria	1,446	44	3.04%	1.3%
Switzerland	3,350	101	3.01%	3.0%
France	5,904	139	2.35%	4.1%
Denmark	2,210	47	2.13%	1.4%
Slovenia	197	4	2.03%	0.1%
Iceland	103	2	1.94%	0.1%
Hungary	221	4	1.81%	0.1%
United Kingdom	35,189	636	1.81%	18.7%
Norway	1,938	35	1.81%	1.0%
Ireland	2,960	51	1.72%	1.5%
Singapore	591	10	1.69%	0.3%
Czechia	599	10	1.67%	0.3%
Canada	10,854	181	1.67%	5.3%
Sweden	2,836	47	1.66%	1.4%
United States	66,207	1,043	1.58%	30.7%
Israel	1,172	18	1.54%	0.5%
Australia	11,390	174	1.53%	5.1%
Belgium	2,538	37	1.46%	1.1%

Possible explanations

The regional pattern is descriptive and we can only speculate about potential causes. Several factors might have favoured uptake in German-speaking Europe and neighbouring countries: Mastodon was developed in Germany and is operated by a German non-profit, and the platform was discussed more prominently in German-language media during the 2022 migration; data protection and the independence from commercial platforms are salient concerns in these countries; and several of the early academic Mastodon instances were run by institutions or communities in German-speaking Europe. However, investigating the driving factors is beyond the scope of this study.

Supplementary Appendix 7: Context of the open science terms in the profile descriptions

Open science endorsement was flagged when a researcher's profile description contained one of the terms listed in Section 2.3 (or a spelling variant). To document the face validity of this flag, Table SA7.1 reports, for the 1,350 flagged researchers, how many matched each term, how many were flagged by that term alone, and which words most often appear immediately before and after the match. Descriptions are not reproduced; the overview is limited to aggregate counts. The majority of flags rest on the term "open science" itself, typically in self-descriptions such as "open science advocate" or "open science enthusiast", whereas "open source" most often refers to software and "open data" occasionally to "linked open data", so a minority of flags may reflect technical practice rather than an open science stance.

Table SA7.1. Terms matched in the profile descriptions of researchers flagged as endorsing open science (N = 1,350)

Term group	People	Flagged by this term only	Most frequent words before	Most frequent words after
open science	797	684	interested (13), rstats (12), researcher (12), data (12), psychology (10)	advocate (40), enthusiast (27), open (18), openaccess (14), rstats (12)
open source	244	203	science (11), free (5), developing (5), openscience (4), biology (3)	software (33), open (11), enthusiast (6), tools (6), data (6)
open access	201	146	openscience (14), interested (4), science (4), data (3), libraries (3)	open (14), openscience (6), opendata (6), advocate (5), journal (5)
open data	153	109	linked (10), openscience (7), openaccess (6), source (4), science (4)	open (12), openscience (6), opengov (3), advocate (3), science (3)
reproducibility / reproducible	103	74	open (11), science (8), openscience (5), about (3)	research (20), science (11), open (6), network (4)
preregistration	3	2		

Note. Spelling variants (e.g., #OpenScience, open-science) are merged. 1,218 researchers matched one term group, 116 two, and 16 three or more. Neighbouring words are the nearest non-function word within three words of the match; counts are numbers of researchers.

Supplementary Appendix 8: Return to Twitter after the migration period

To assess whether the observed migration persisted, we compared the Twitter and Mastodon activity of migrated researchers in April 2023, four months after the migration period. For each migrated researcher, we counted tweets and Mastodon posts between 1 April

and 30 April 2023 (UTC). A researcher was classified as having returned to Twitter if their Twitter activity in April 2023 exceeded their Mastodon activity in that month. Of the 3,990 migrated researchers, 3,361 (84.2%) indeed had higher activity on Twitter. This indicates that for most researchers the shift to Mastodon was temporary. The return flag is included in the deposited data set (column is_returned).

References

Bosnjak, M., Fiebach, C. J., Mellor, D., Mueller, S., O'Connor, D. B., Oswald, F. L., & Sokol, R. I. (2022). A template for preregistration of quantitative research in psychology: Report of the joint psychological societies preregistration task force. *American Psychologist*, *77*(4), 602–615. https://doi.org/10.1037/amp0000879

Japkowicz, N. (2000). The class imbalance problem: Significance and strategies. *Proc. Of the Int'l Conf. On Artificial Intelligence*, *56*, 111–117.

King, G., & Zeng, L. (2001). Logistic regression in rare events data. *Political analysis, 9*(2), 137-163.

La Cava, L., Aiello, L. M., & Tagarelli, A. (2023). Drivers of social influence in the Twitter migration to Mastodon. *Scientific Reports*, *13*(1). https://doi.org/10.1038/s41598-023-48200-7

Mongeon, P., Bowman, T. D., & Costas, R. (2023). An open data set of scholars on Twitter. *Quantitative Science Studies, 4*(2), 314–324. https://doi.org/10.1162/qss_a_00250

Mongeon, P., Bowman, T. D., Costas, R., & Arroyo-Machado, W. (2024). Open dataset of scholars on Twitter (X) (Version 2) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.10905839

Müller, S. M., Kotzur, M., Bittermann, A. (2023). *An Approach for Researcher Identification on Twitter Without the Need for External Data.* PsychArchives. https://doi.org/10.23668/psycharchives.13254

Van Eck, N. J. (2024). *Classification of research publications based on data from OpenAlex* (Version 2023nov) [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.10560276

Willroth, E. C., & Atherton, O. E. (2024). Best laid Plans: A guide to reporting preregistration deviations. *Advances in Methods and Practices in Psychological Science*, *7*(1). https://doi.org/10.1177/25152459231213802